

A 附录

B 带节点和边约束的 EOFT 算法细节

Algorithm 2 OFT-Sinkhorn: Iterative Sinkhorn-based Algorithm for (capacited) Entropic Optimal Flow Transport with Edge and Node Constraints

Input: Distance Matrix $\mathbf{D}$, Edge Capability $\mathbf{S}$, Node Capability $\mathbf{r}$, Marginal Difference $\mathbf{s}$, maximum iteration number L , Virtual-flow $\mathbf{d}$.

Output: The Allocation Matrix $\mathbf{P}_{ij}$ for $i, j = 1, \dots, N$

Initialize $\mathbf{K} = \exp\left(-\frac{\mathbf{D}}{\epsilon}\right)$ element-wise.

for $l = 0$ **to** $L - 1$ **do**

 Update $\mathbf{u}^{(l+1)}, \mathbf{v}^{(l+1)}$ according to Eq. 8

 Update $\mathbf{q}^{(l+1)}$ according to Eq. 13

 Update $\mathbf{G}^{(l+1)}$ according to Eq. 14

 Update $\mathbf{K}^{(l+1)}$ by $\exp\left(\frac{-\mathbf{D} + \mathbf{G}^{(l+1)}}{\epsilon}\right)$

end for

Compute $\mathbf{P} = \text{Diag}(\mathbf{u}) \cdot \mathbf{K} \cdot \text{Diag}(\mathbf{v})$

return $\mathbf{P}^* = \mathbf{P} - \mathbf{P}^\top$

算法 EOFT-SH 与 EOFT-SH-C 的证明

此处给出正文中命题 1 与命题 2 的数学证明。

C.1 无约束情形

$$\begin{aligned} \min_{\substack{\mathbf{P} \geq 0, \\ \mathbf{q} \geq 0}} \quad & \langle \mathbf{D}, \mathbf{P} \rangle - \epsilon H(\mathbf{P}) \\ \text{subject to} \quad & \mathbf{P} \mathbf{1}_N = \mathbf{q} + \mathbf{d}, \mathbf{P}^\top \mathbf{1}_N = \mathbf{q} - \mathbf{s} + \mathbf{d} \\ & \text{Diag}(\mathbf{P}) = \mathbf{d} \end{aligned} \quad (16)$$

引入对偶变量 $\mathbf{f}, \mathbf{g}, \mathbf{h}$ 后, 拉格朗日方程为:

$$\mathcal{L} = \langle \mathbf{D}, \mathbf{P} \rangle - \epsilon H(\mathbf{P}) - \langle \mathbf{f}, \mathbf{P} \mathbf{1}_N - \mathbf{q} - \mathbf{d} \rangle - \langle \mathbf{g}, \mathbf{P}^\top \mathbf{1}_N - \mathbf{q} + \mathbf{s} - \mathbf{d} \rangle - \langle \mathbf{h}, \text{Diag}(\mathbf{P}) - \mathbf{d} \rangle \quad (17)$$

对拉格朗日方程关于 $\mathbf{P}$ 求偏导可得:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{P}_{ij}} = \mathbf{D}_{ij} + \epsilon \log(\mathbf{P}_{ij}) - \mathbf{f}_i - \mathbf{g}_j = 0 (i \neq j) \quad (18)$$

进而可得:

$$\mathbf{P}_{ij} = e^{\mathbf{f}_i/\epsilon} e^{-\mathbf{D}_{ij}/\epsilon} e^{\mathbf{g}_j/\epsilon}. \quad (19)$$

类似地:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{P}_{ii}} = \mathbf{D}_{ii} + \epsilon \log(\mathbf{P}_{ii}) - \mathbf{f}_i - \mathbf{g}_i - \mathbf{h}_i = 0 \quad (20)$$

进而可得:

$$\mathbf{P}_{ii} = e^{\mathbf{f}_i/\epsilon} e^{-(\mathbf{D}_{ii} - \mathbf{h}_i)/\epsilon} e^{\mathbf{g}_i/\epsilon}. \quad (21)$$

因此, 对于整个矩阵 $\mathbf{P}$, 可得:

$$\mathbf{P} = \text{Diag}(e^{\mathbf{f}/\epsilon}) e^{-\mathbf{D} + \text{Diag}(\mathbf{h})/\epsilon} \text{Diag}(e^{\mathbf{g}/\epsilon}). \quad (22)$$

令 $\mathbf{u} = e^{\mathbf{f}/\epsilon}$, $\mathbf{v} = e^{\mathbf{g}/\epsilon}$, $\mathbf{K} = e^{-\mathbf{D} + \text{Diag}(\mathbf{h})/\epsilon}$, 可从 $\mathbf{P} =$ 得到最优解 $\text{Diag}(\mathbf{u}) \mathbf{K} \text{Diag}(\mathbf{v})$.